Introduction to PHP

- PHP stands for Hypertext Preprocessor
- It is a pgmming lang used to create dynamic and interactive web pages.
- open source
- Server side scripting language
- All PHP code must be enclosed within

```
<?php
. . . .
. . . .
?>
```
tags
- Every PHP stmt must end in a semicolon
- Sample pgm

```
<?php
echo "Hello world<br>";
echo "How are you";
?>
```

Variables :- used to store info like numbers, text etc. Variable names are case-sensitive but keywords aren't.

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They are declared using $ symbol.

$x =10
$y = "Hello world"

- constants - to store values that cannot be changed during execution

Comments - unexecutable code used for documentation
→ single line must be preceded by the // characters, while multiline comments must be enclosed within a /* .... */ comment block

Strings
$name = "Hello world"
echo strlen($name); → o/p : 11
echo str-word-count($name); → o/p : 2
echo strrev($name); → o/p : dlrow olleH
echo strtolower($name); → o/p : hello world
echo strpos($name, 'world!' 'You') → o/p : 6
echo str-replace('Hello' 'Are', 'Hi', $name); → o/p : ~~Hello~~ Hi world

Math functions
echo pi; → 3.14
echo sqrt(16); → 4
echo abs(-20); → 20

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echo min(0,3,4,8); → 0
echo max(0,3,4,8); → 8
echo round(1.7); → 2
echo rand();
echo rand(1000,9999);

Operators
- Arithmetic operators → +, -, *, /, %
- Relational operators → <, >, <=, >=, ==, ===, !=, !==
  $a=10;
  $b='10';
  ~~echo $~~
  if $a = = $b true
  if $a = = = $b false
  ↳ checks data type also
- Increment & decrement operators → ++ & --
  $a++ → post increment
  ↳ return value of $a, then increments its value by one
  ++$a → pre increment
  ↳ increments value of $a by one, then returns value of $a

conditional stmt.
if ($a = = 5)
echo "num is 5";
else
echo "some other num";

if ($a==10)
{ $a=5;
echo "Hello";
}
else.

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```
• $a=17;
  switch($a)
   { case 10:
       echo "Hi";
        break;
     case 17:
       echo "Hello";
        break;
     default:
       echo "No";
   }
```

Loops

• for loop

```
for($x=1;$x<=10;$x++)
{echo "this number is $x <br>";
}
```

• while loop

```
$x=1;
while($x<=10)
{echo "this num is $x <br>";
$x++;
}
```

• do while

```
$x=1;
do
{echo "this num is $x";
 $x++;
}
while($x<=10);
```

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Jumping stmts

```
• for($x=1;$x<=10;$x++)
  { if($x==4)
      continue;
      echo "this num is $x";
   }
```

```
• for($x=1;$x<=10;$x++)
  { if($x==4)
      break;
     echo "this num is $x";
   }
```

Functions

```
Add two(){
    $a=5;
    $b=10;
    echo $a+$b;
 }
```

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5. write a pgm to check whether the given no is even or odd

Ans
```
<?php
$n=10;
if ($n%2==0)
  echo "even no";
else
  echo "odd no";
?>
```

5. wap to find factorial of a number
```
<?php
$n=3;
$f=1;
for($i=1;$i<=3;$i++)
{ $f=$f*$i; }
if ($n==0)
  echo "factorial is 1";
else
  echo "factorial is $f";
?>
```

6. wap to create a calculator using PHP

Ans
```
<?php
$a=10;
$b=5;
```

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```
$op='+';
switch ($op)
case '+':
    $add=$a+$b;
    echo "result is $add";
    break;
case '-';
    $sub=$a-$b
    echo "res is $sub";
    break;
case '*';
    $mul=$a*$b
    echo "res is $mul";
    break;
case '/';
    if ($b==0)
     echo "divide by zero";
    else
     $div=$a/$b;
     echo "result is $div";
     break;
case %;
     $mod=$a%$b
      echo "res is $mod";
      break;
default: echo "invalid input";
    ?>
```

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Get and Post

They are the methods used to pass info from one page to another.

Get	Post
• Info is visible to everyone	not visible to everyone
• can send only limited amt of info (approx 2k characters)	8mb info can be sent
• not used to send sensitive info	used to pass sensitive info
• not suitable for large amt of info	used for passing large amt of info
• we can bookmark the info (Since its visible in titlebar)	can't bookmark (details will be hidden)

Q. check whether the number is prime or not.

```
<?php
$a=3;
$count=0;
for($i=1;$i<=$a;$i++)
   if $i%$a==0
      $count+=1;
   if $count==2
      echo "prime"
   else
      echo "not prime"
?>
```

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Q. Find the sum of array elements

```
<?php
$arr=[10,12,6,14,18,20];
$sum=array_sum($arr);
  echo($sum);
?>
```

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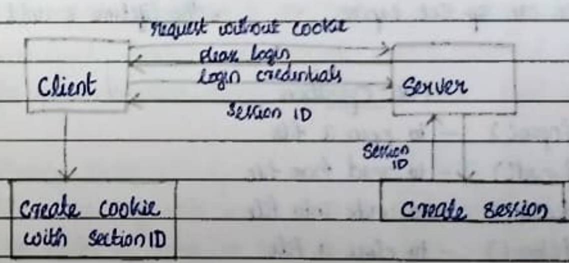

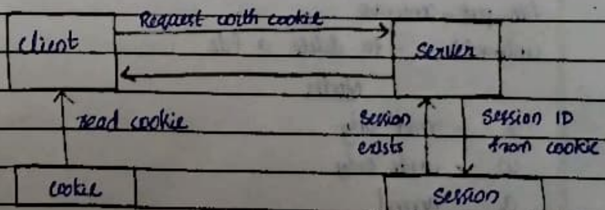

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Session	cookie
* It stores user data on the server and uses the unique ID to access it	* Is a small piece of data stored in the user's browser. It is sent to every request with the server
* session size is large	* cookie size is small
* more secure	* less secure
* we can ~~sp~~ set expire	* The lifetime is until browser closes

File operations

fopen() - to open a file
fread() - to read from file
fwrite() - to write into file
fclose() - to close a file
file_exists()
file_get_contents
file_put_contents
unlink() - to delete a file

Modes

r - read only
w - write only
a - append
r+ - read and write
w+ - read and write
a+ - read and append

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Errors & Exception Handling

1. Parse Error (Syntax Error)
It's a syntax error done by the programmer in the source code. It will be caught by the compiler. It is caused due to unclosed quotes, missing or extra parenthesis, missing semicolon etc.

2. Fatal Error
Php compiler understands the code but it recognises an undeclared function ie function called without definition. The pgm execution stops when there is a fatal error.

3. Warning Error
It's a missing file error. but it won't terminate the script
eg: failed to open, no such file etc.

4. Notice Error
It's similar to warning error pgm contains something wrong but allows it's execution.

Exception handling

try
catch
throw
finally